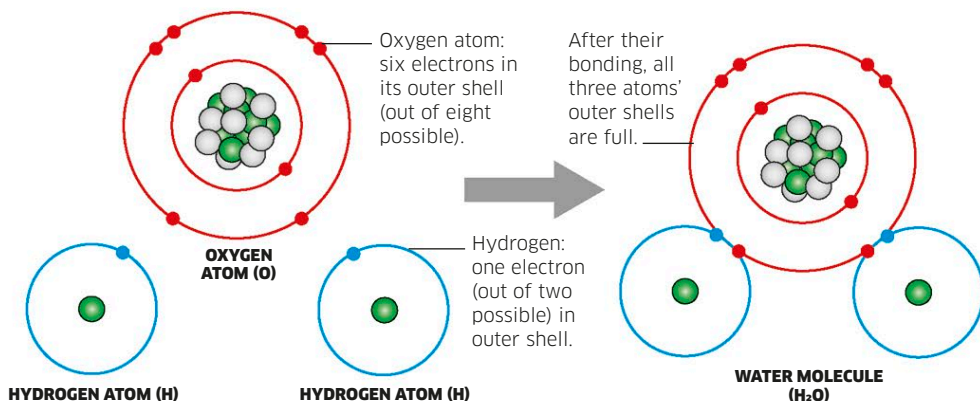
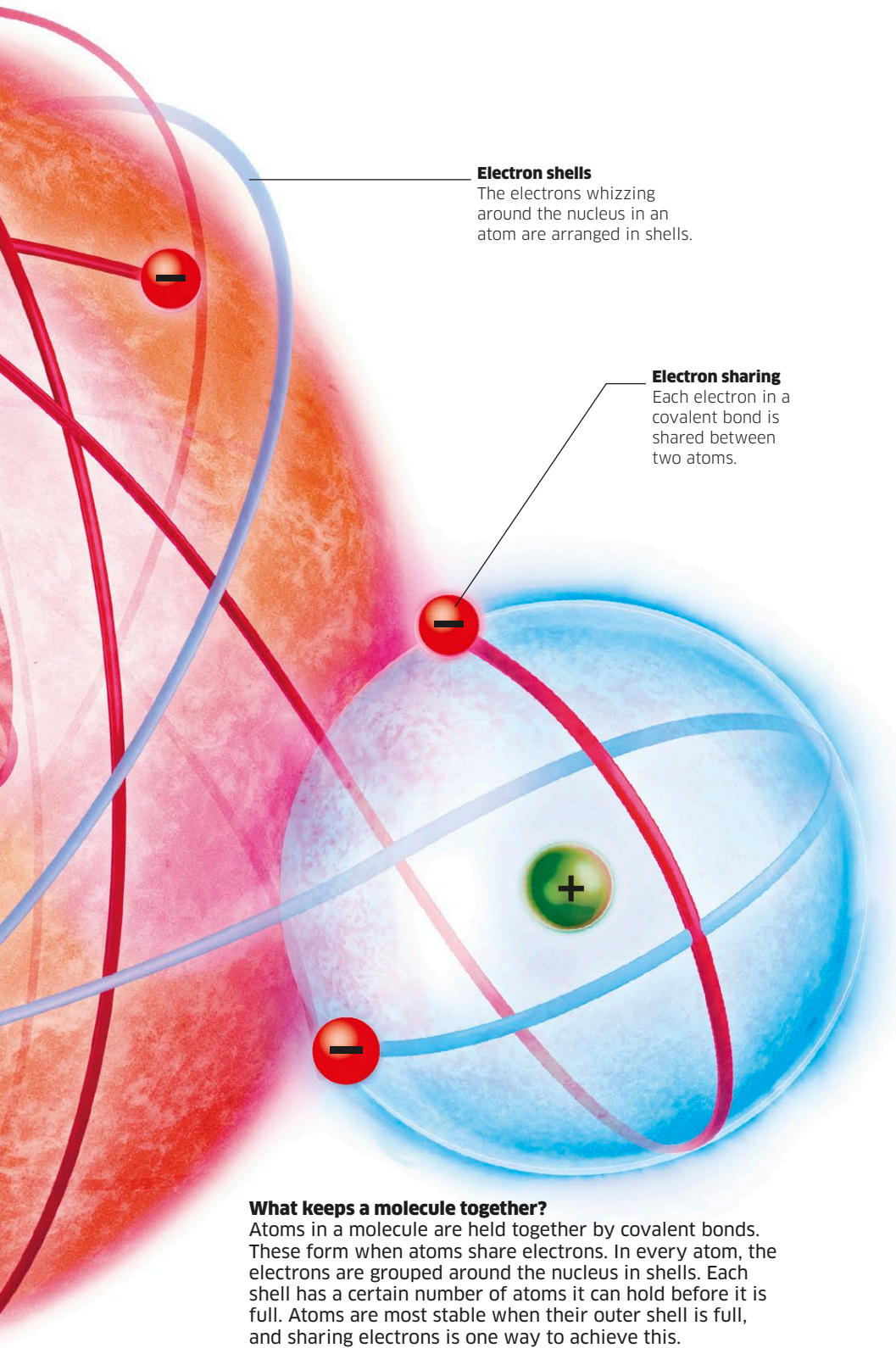


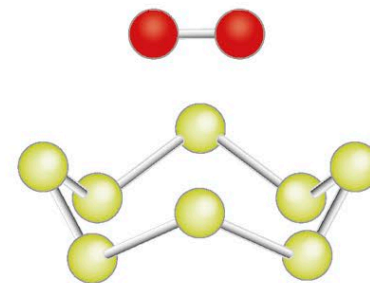


Elements and compounds

Most elements are made up of single atoms, but some are made of molecules of two or more identical atoms. When two elements react, their molecules form a new compound.

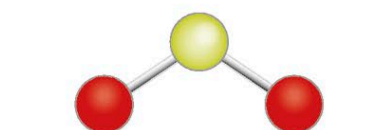
Oxygen

The gas oxygen (O₂) is made of molecules, each containing two oxygen atoms.



Sulfur

Pure sulfur (S), a solid, normally exists as molecules of eight sulfur atoms bonded together.

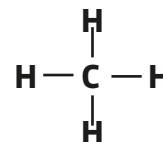


Sulfur dioxide (SO₂)

When sulfur and oxygen molecules react, their bonds break to make new bonds and a new substance forms.

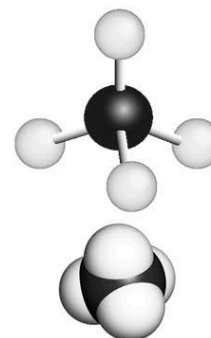
Representing molecules

Scientists have different ways of representing molecules to understand how chemical reactions happen. Here, a molecule of the gas compound methane (CH₄), made of one carbon atom and four hydrogen atoms, is shown in three ways.



Lewis structure

The simplest way to represent a molecule is to use the chemical symbols (letters) and lines for covalent bonds.



Ball and stick

Showing the atoms as balls and the bonds as sticks gives a three-dimensional representation of a molecule.

Space filling

This method is used when the space and shape of merged atoms in a molecule are more important to show than bonds.

Macromolecules

While some compounds are made of small molecules consisting of just a few atoms, there are many compounds whose molecules are made of thousands of atoms. This molecular model shows a single molecule of a protein found in blood, called albumin. It contains atoms of many different elements, including oxygen, carbon, hydrogen, nitrogen, and sulfur.

